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MTH202 (Mega File)

Mid Term (Live Quiz)

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1. The matrix representation of an inverse relation can be obtained by taking the of the relational matrix.
 - inverse
 - transpose ☒
 - adjoint
 - Determinant
2. Let $R = \{(1, 2), (3, 4), (5, 6), (7, 8)\}$. Range of the inverse of the relation is
 - $\{2, 4, 6, 8\}$
 - $\{5, 6, 7, 8\}$
 - $\{1, 2, 3, 4\}$
 - $\{1, 3, 5, 7\}$ ☒
3. What will be the fourth term of the following sequence $1/2, 2/3, 3/4$?
 - $6/7$
 - $5/6$
 - $4/5$ ☒
4. Operation of subtraction is a binary operation on the set of
 - Natural numbers
 - Positive integers



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- Integers ☒
 - Whole numbers
5. Let $f(x) = 2x$ and $g(x) = x + 2$ define functions f and g from $\mathbb{R}$ to $\mathbb{R}$, then $(f - g)(x)$ is
- $2x^2 + 4x$
 - $2x + 2$
 - $3x + 2$
 - $x - 2$ ☒
6. Common ratio in sequence "36, 12, 4, 4/3....." is
- 4
 - 3 ☒
 - 2
 - 1
7. Let f and g be the two functions from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = |x|$ and $g(x) = \sqrt{x}$ for all $x \in \mathbb{R}$. Then
- $f(x) = g(x)$ ☒
 - $f(x) \neq g(x)$
 - $f(x) < g(x)$
 - $f(x) > g(x)$
8. $R = \{(a, 1), (b, 2), (c, 3), (d, 4)\}$ then the inverse of this relation is
- $\{(a, 1), (b, 2), (3, c), (4, d)\}$ ☒
 - $\{(1, a), (b, 2), (3, c), (4, d)\}$
 - $\{(1, a), (2, b), (3, c), (4, d)\}$
 - $\{(a, 1), (2, b), (3, c), (4, 4)\}$
9. Consider a function f defined on $X = \{1, 2, 3, 4\}$ by $f(1) = 1$, $f(2) = 2$, $f(3) = 3$ and $f(4) = 4$ then
- f is an identity function ☒



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- f is a constant function

10. Cardinality of positive prime numbers less than 20 is

- 10
- 20
- 8 ☒
- 5

Let $R = \{(1, 2), (3, 4), (5, 6), (7, 8)\}$. Domain of the inverse of the relation is _____.

- $\{1, 3, 5, 7\}$
- $\{5, 6, 7, 8\}$
- $\{2, 4, 6, 8\}$ ☒
- $\{1, 2, 3, 4\}$

2. If f is a bijective function then $(f^{-1}(f(x)))$ is equal to:

- None of these ☒
- $f(x)$
- $f^{-1}(x)$

3. A set is called finite if, and only if, it is the:

- empty set or onto ☒
- one-to-one or onto
- empty set or bijective
- empty set or one-to-one



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4. Let $A = \{1, 2, 3\}$ and R be the relation defined on A such that $R = \{(1, 1), (1, 2), (2, 3), (3, 1)\}$. Then R is _____.

- Antisymmetric
- symmetric ☒
- neither symmetric nor antisymmetric
- reflexive

5. Why the function $f : \mathbb{Z} \rightarrow \mathbb{R}$ defined by $f(x) = 1/(x^2 - 25)$ is not well defined?

- Because f is not defined for $x > 0$
- Because f is not defined for $x < 0$
- Because f is not defined for 5 and -5 ☒
- Because f gives imaginary values for all integers

6. Let $A = \{1, 2, 3, 4\}$ and $B = \{7\}$, then the constant function from A to B is:

- Both one to one and onto
- Neither one to one nor onto ☒
- One to one
- Onto

7. Let $R = \{(a, 1), (b, 2), (c, 3), (d, 4)\}$ then the inverse of this relation is:

- $\{(1, a), (2, b), (3, c), (4, d)\}$ ☒
- $\{(1, a), (2, b), (3, c), (4, d)\}$
- $\{(a, 1), (2, b), (3, c), (4, d)\}$
- $\{(a, 1), (b, 2), (3, c), (4, d)\}$



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8. Let $A = \{1, 2, 3, 4\}$ and define the relation R on A by $R = \{(1, 2), (2, 3), (3, 3), (3, 4)\}$. Then R is _____.

- Reflexive
- Neither reflexive nor irreflexive ☒
- Irreflexive
- Both reflexive and irreflexive

9. Let $f(x) = 2x$ and $g(x) = x + 2$ define functions f and g from R to R , then $(f - g)(x)$ is:

- $x - 2$ ☒
- $3x + 2$
- $2x + 2$
- $2x^2 + 4x$

1. Let $A = \{1, 2, 3, \dots, 50\}$ and $B = \{2, 4, 6, 8, 10\}$. Then the Cartesian product of A and B has elements.

- ☐ 100
- ☐ 250
- ☒ 50

2. The relation $R = \{(2, 4), (x, 1), (4, 2), (5, 6)\}$ is a function if x is not equal to

- ☐ 2, 4 or 6
- ☒ 1, 2 or 6
- ☐ 4, 1 or 6

3. Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$ and let R be the "divides" relation from A to B i.e., for all $(a, b) \in A \times B$, $a R b$ iff $a \mid b$ (a divides b). Then



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- ☐ $R = \{(6,8), (6,8)\}$
- ☐ $R = \{(2,6), (3,8), (4,8)\}$
- ☒ $R = \{(2,2), (2,6), (2,8), (3,6), (4,8)\}$
- ☐ $R = \{(2,2), (2,6), (2,8), (3,6), (4,8), (6,8)\}$

4. Input values of the function are called the

- ☐ Relation
- ☐ Range
- ☒ Domain
- ☐ None of these

5. Which of the following is always true for the matrix representation of a symmetric relation?

- ☒ Matrix is equal to its transpose.
- ☐ Matrix has always 1 in its diagonal entries.
- ☐ Matrix is singular.
- ☐ Matrix has always 0 in its diagonal entries.

6. Cartesian product of two sets symbolically written as

- ☐ $A \times B = \{(a,b) \mid a \in A \text{ and } b \in B\}$
- ☐ $AB = \{(a,b) \mid a \in A \text{ and } b \in B\}$
- ☐ $A \times B = \{(a,b) \mid a \in A \text{ and } b \in B\}$
- ☒ $A \times B = \{(a,b) \mid a \in A \text{ and } b \in B\}$

7. Let R be the universal relation on a set A , then which one of the following statements about R is true?



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- ☐ R is not reflexive.
- ☒ R is reflexive, symmetric and transitive.
- ☐ R is not symmetric.
- ☐ R is not transitive.

8. Let $A = \{1, 2, 3\}$ and R be the relation defined on A such that $R = \{(1,1), (1,2), (2,3), (3,1)\}$, then R is _____.

- ☒ Symmetric
- ☐ Antisymmetric

9. Logic gate NOT does not define a binary operation on $\{0,1\}$ because.....

- ☐ It takes two inputs and gives multiple outputs.
- ☐ It takes a single input and gives more than one output.
- ☒ It takes a single input and gives a single output.
- ☐ It takes two inputs and gives a single output.

10. There is at least one loop in the graph of an irreflexive relation.

- ☐ True
- ☒ False

❓ Composite Relation symbolically written as:

- $SOR = \{(a,c) \mid a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$
- $SOR = \{(a,c) \mid a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$ ☒
- None of the above
- $SoR = \{(a,c) \mid a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$



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❑ Subtraction (-) is a binary operation on the set of integers.

- False ☒
- True

❑ Let the set $A = \{1, 2, 3, 4\}$. Then the relation $\{(2, 4), (4, 2)\}$ is:

- Reflexive
- Transitive
- Symmetric ☒
- Antisymmetric

❑ Let $A = \{1, 2, 3, 4\}$ and define relation $R = \{(1, 1), (1, 3), (2, 4), (3, 1), (4, 2)\}$ on A . Then which of the following statement about R is true?

- R is symmetric
- R is transitive
- R is reflexive
- R is not symmetric ☒

❑ $25 = 1 \pmod{3}$ means that 3 divides:

- $25 - 1$ ☒
- $25 \div 71$
- $25 + 1$

❑ The matrix representation of an inverse relation can be obtained by taking the:

- Determinant
- Adjoint
- Inverse
- Transpose ☒

❑ Which of the following relations are functions?

- $R_1 = \{(3, 4), (4, 5), (6, 7), (8, 9)\}$



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- $R_2 = \{(3, 4), (4, 5), (6, 7), (3, 9)\}$
- $R_3 = \{(-3, 4), (4, -5), (0, 0), (8, 9)\}$
- $R_4 = \{(8, 11), (3, 4), (5, 6), (17, 8), (19, 9)\}$ ✓

❓ Which of the following relations are functions?

- $R_1 = \{(3, 4), (4, 5), (6, 7), (8, 9)\}$
- $R_2 = \{(3, 4), (4, 5), (6, 7), (3, 9)\}$
- $R_3 = \{(-3, 4), (4, -5), (0, 0), (8, 9)\}$
- $R_4 = \{(8, 11), (3, 4), (5, 6), (17, 8), (19, 9)\}$ ✓

❓ The number of elements in the power set of $\emptyset$ denoted by $P(\emptyset)$ is:

- 2^0 ✓
- n
- 2^2

❓ Suppose A has 3 elements and B has 4 elements, then the number of the binary relations from A to B that are functions is:

- $2^6 / 2^8$
- $2^6 / 2^{12}$
- $2^1 / 2^6$
- $2^6 / 2^4$ ✓

. Logic gate NOT does not define a binary operation on $\{0,1\}$ because:

- It takes a single input and gives a single output ✓
- It takes two inputs and gives a single output.
- It takes a single input and gives more than one output.
- It takes two inputs and gives multiple outputs.



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2. Let R and S be reflexive relations on a set A then $R \cap S$ is reflexive.

- True ✓
- False

3. Let $R = \{(1, 2), (3, 4), (5, 6), (7, 8)\}$. Domain of the inverse of the relation is:

- $\{1, 3, 5, 7\}$
- $\{5, 6, 7, 8\}$
- $\{2, 4, 6, 8\}$ ✓
- $\{1, 2, 3, 4\}$

4. If a set A has 15 elements then $P(A)$ (power set of A) has _____ elements.

- 2^{16}
- 2^{15}
- 2^{14}
- 2^{13} ✓

5. The matrix representation of an inverse relation can be obtained by taking the _____ of the relational matrix.

- adjoint
- Determinant
- inverse
- transpose ✓

6. Select the correct option for:

- Set of positive Real numbers



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- Set of integers
- Set of negative Real numbers
- Set of Real numbers ✓

7. Range of function $f(x) = e^{Ax}$ is:

- Set of positive Real numbers ✓
- Set of integers
- Set of negative Real numbers
- Set of Real numbers

8. Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$ and let R be the "divides" relation from A to B i.e., for all $(a, b) \in A \times B$, $a R b$ iff a divides b . Then:

- $R = \{(6), (a)\}$
- $R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8), (6, 8)\}$
- $R = \{(2, 6), (3, 6), (4, 8)\}$
- $R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8)\}$ ✓

9. Let R be a relation on a set A . R is transitive if and only if for all $a, b, c \in A$ then:

- $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$ ✓
- $(a, b) \in R$ then $(b, a) \in R$
- $(a, b) \in R$ and $(b, c) \in R$ but $(a, c) \notin R$
- $(a, b) \in R$ then $(b, a) \notin R$

10. Let $A = \{1, 2, 3\}$ and $B = \{2, 4\}$ then the number of binary relations from A to B are:

- 64 ✓



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- 32
- 128
- 256

11. Which of the following is always true for the matrix representation of a symmetric relation?

- Matrix has always 1 in its diagonal entries
- Matrix has always 0 in its diagonal entries
- Matrix is singular
- Matrix is equal to its transpose ✓

1. If $A = \{1, 2, 3\}$ is a set and $R = \{(1, 2), (2, 2), (2, 1)\}$ is a relation on A, R is:

- ☐ Transitive
- ☐ None of these
- ☒ Symmetric

2. Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$ and let R be the "divides" relation from A to B. Then:

- ☐ $R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8)\}$
- ☒ $R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8), (6, 8)\}$
- ☐ $R = \{(2, 6), (3, 6), (4, 8)\}$
- ☐ $R = \{(3, 6), (3, 8)\}$

3. In the representation of an irreflexive relation using a matrix, all its diagonal entries will be:

- ☒ 0
- ☐ 1

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4. The domain of a relation R from A to B is symbolically written as:

- ☒ $D(R) = \{a \text{ belongs to } A \text{ such that } (a,b) \text{ belongs to } R\}$
- ☐ $D(R) = \{a \text{ belongs to } B \text{ such that } (a,b) \text{ belongs to } R\}$

5. If $x = -10 \pmod{15}$, which of the following integers are valid solutions for x ?

- ☐ 6
- ☒ 6

6. Let $A = \{1,2,3,4\}$ and define the relation R on A by $R = \{(1,2), (2,3), (3,3), (3,4)\}$. Then:

- ☐ R is reflexive
- ☐ R is both reflexive and irreflexive
- ☒ R is neither reflexive nor irreflexive
- ☐ R is irreflexive

7. Why is the function $f : \mathbb{Z} \rightarrow \mathbb{R}$ defined by $f(x) = \sqrt{x}$ not well defined?

- ☐ Because f is not defined for 5 and -5.
- ☐ Because f is not defined for $z < 0$.
- ☐ Because f is not defined for $z > 0$.
- ☒ Because f gives imaginary values for all integers.

8. Operation of subtraction is a binary operation on the set of:

- ☐ Natural numbers
- ☐ Whole numbers
- ☒ Integers



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- ☐ Positive integers

9. If a relation $R = \{(1,1), (2,1), (1,2), (2,2)\}$ is given, then which of the following is not true about this relation?

- ☐ R is Symmetric.
- ☒ R is Irreflexive.
- ☐ R is Reflexive.
- ☐ R is Transitive.

10. The logic gate NOT is a unary operation on $\{0, 1\}$:

- ☐ False
- ☒ True

1. If $f(x) = 4x - 8$, find $f(5)$

- ☐ 12
- ☒ 14
- ☐ 18
- ☐ 15

2. The inverse relation R^{-1} from B to A is defined as _____.

- ☒ $R^{-1} = \{(b,a) \in B \times A \mid (a,b) \in R\}$
- ☐ None of the above
- ☐ $R^{-1} = \{(b,a) \notin B \times A \mid (a,b) \in R\}$
- ☐ $R^{-1} = \{(b,a) \in B \times A \mid (b,a) \in R\}$



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3. The matrix representation of an inverse relation can be obtained by taking the _____ of the relational matrix.

- ☒ transpose
- ☐ inverse
- ☐ determinant
- ☐ adjoint

4. Given a Set $A=\{a,b\}$ and $B=\{1,2,3\}$, then number of functions from A to B is....

- ☒ 9
- ☐ 12
- ☐ 6
- ☐ 18

5. Range of the relation $\{(0,1), (3,22), (90,34)\}$ is

- ☐ $\{0,1,3,22,90,34\}$
- ☐ $\{0,3,90\}$
- ☒ $\{1, 22, 34\}$
- ☐ $\{0,13\}$

6. The number of elements in $A \times B$ are _____ if A is a set with '5' elements and B is a set with '4' elements.

- ☒ 20
- ☐ 10
- ☐ 15
- ☐ 25



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7. The Range of $f: X \rightarrow Y$ is also called the image of _____ under f .

- ☒ True
- ☐ False

8. Let the set $A = \{1,2,3,4\}$ Then the relation $\{(2,4), (4,2)\}$ is

- ☐ Reflexive
- ☒ symmetric
- ☐ Transitive
- ☐ AntiSymmetric

9. If $A = \{1,2,3\}$ & $B = \{4,5,6\}$ and $R = \{(1,4), (2,5), (3,6), (3,4)\}$ The complementary relation is

- ☐ $A \times B$ (intersection) R
- ☒ $A \times B$
- ☐ $A \times B$ (Union) R
- ☒ $A \times B$ (difference or $-$) R

10. Range of a relation symbolically written as

- ☒ $\text{Ran}(R) = \{b \in B \mid (a,b) \in R\}$
- ☐ None of the above
- ☐ $\text{Ran}(R) = \{a \in A \mid (a,b) \in R\}$
- ☐ $\text{Ran}(R) = \{b \in B \mid (a,b) \in R\}$

Range of function $f(x) = |x|$ will be:

- Set of negative real numbers



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- Set of whole numbers
- Set of positive real numbers ✓
- Set of real numbers

☐ Let $A = \{1, 2, 3, \dots, 50\}$ and $B = \{2, 4, 6, 8, 10\}$. Then the Cartesian product of A and B has:

- 100 ✓
- 50
- 250

☐ Subtraction (-) is a binary operation on the set of integers:

- False
- True ✓

☐ Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, then which of the following sets represent the partition of the set A?

- $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 4, 6\}$, $C = \{8\}$ ✓
- $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 3, 4, 6\}$, $C = \{8\}$
- $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 4, 6\}$, $C = \{8, 9\}$

☐ Let $S = R$ and define the 'square' relation $R = \{(x, y) | z^2 = y\}$. The square relation is an:

- Not equivalence relation
- Equivalence relation ✓

☐ Suppose A has 3 elements and B has 4 elements, then the number of functions from A to B are:

- 81 ✓
- 12
- 8

☐ Let $P(A)$ be the power set of a non-empty set A and by using the definition of "subset" relation, c , for all $X, Y \in P(A)$, $X \subset Y$ if $x \in X$ then $x \in Y$. Then:

- c is a partial order relation. ✓
- c is not a partial order relation.



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☐ The range of a function is alwaysthe co-domain of f:

- a superset of
- equal to
- a subset of ✓
- not equal to

☐ The relation $R = \{(12, 14), (13, 5), (-2, 7), (x, 13)\}$ is a function if x is not equal to:

- 14, 5 or 7. ✓
- 12, 13 or -2.
- 12, 13 or 2.
- 14, 7 or 13.

☐ Which of the following logical connectives is not a binary operation?

- Disjunction
- Negation ✓
- Implication
- Conjunction

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